Timing

File: timing.kicad_sch

Sprite Table

File: sprite_table.kicad_sch

Sprite RAM

Sprite ROM 1

File: sprite_rom_1.kicad_sch

Sprite ROM logic

PWB(C)

Sprite ROM 2

File: sprite_rom_logic.kicad_sch File: sprite_rom_2.kicad_sch

PWB(C) is used when sprite ROMs P13,P16,P15 and P18 on PWB(B) are not mounted.

File: sprite_ram.kicad_sch

Sprite Frame Buffers

File: sprite_frame_buffers.kicad_sch

Spare parts on video PWB(B) 302191A

B_B9C 74LS04 B_B9B 74LS04 B_B9E 74LS04 B_B9D 74LS04 B_P11B 74LS74 B_R11A 74LS32

Spare parts are set as NC.

Spare parts on main PWB(A) 302190A

A_L2A 74LS00 A_L2D 74LS00 A_H8A 74LS07 A_H8D 74LS07 A_K1C 74LS32 A_L1C 74LS07 A_L1B 74LS07 A_L8F 74HC367 A_N3B 74LS139 decoder GND

KiCad does not support reference designators starting with a number. Therefore the references used in the original schematics have been mirrored. 6R -> R6. The original schematics had separate pages for the main and video PCB. They are combined here. This makes some references duplicate. To distinguish between boards A_ is added before part positions on PWB(A) and for PWB(B) B_ and so on. If there are still conflicts a "P" (Position) is added after board letter AP_, BP. Sometimes there are duplicate reference designators then "CAP" maybe added after board letter ACAP_ (CAP is for capacitor).

Tile Layers

File: tile_layers.kicad_sch

Fixed Layer

File: fixed_layer.kicad_sch

IO

File: io.kicad_sch

Color mixer

File: color_mixer.kicad_sch

Sound

File: sound.kicad_sch

Sound mixer

File: sound_mixer.kicad_sch

Timing diagrams

File: timing_diagrams.kicad_sch

Sprite timing diagrams

File: sprite_timing_diagrams.kicad_sch

Main CPU

File: mcpu.kicad_sch

Shared Resources

File: shared_resources.kicad_sch

Slave CPU

File: scpu.kicad_sch

Only used on GX903 – Cue Brick

Battery Powered RAM

File: bpram.kicad_sch

There are several games and variations on the twin 16 platform:
- Main Board PWB(A)
- Video Board PWB(B)
- Additional ROM Board, if not ROM's mounted directly on video board.
- Battery powered RAM board

	Id	PWB(A)	PWB(B)	PWB(C)
Dark Adventure	GX687	302190A	302191A	–
Gradius 2	GX785	302190A	302191A	–
Vulcan Venture	GX785	302190A	302191A	–
Gradius 2	GX785	302190B	302191B	402145A
Hard Puncher	GX870	302190B	302191B	402145A
M.I.A	GX808	302190B	302191B	402145A
Cue Brick	GX903	302190B	302191B	–

Cue Brick has an additional Battery Powered RAM Board: PWB451803

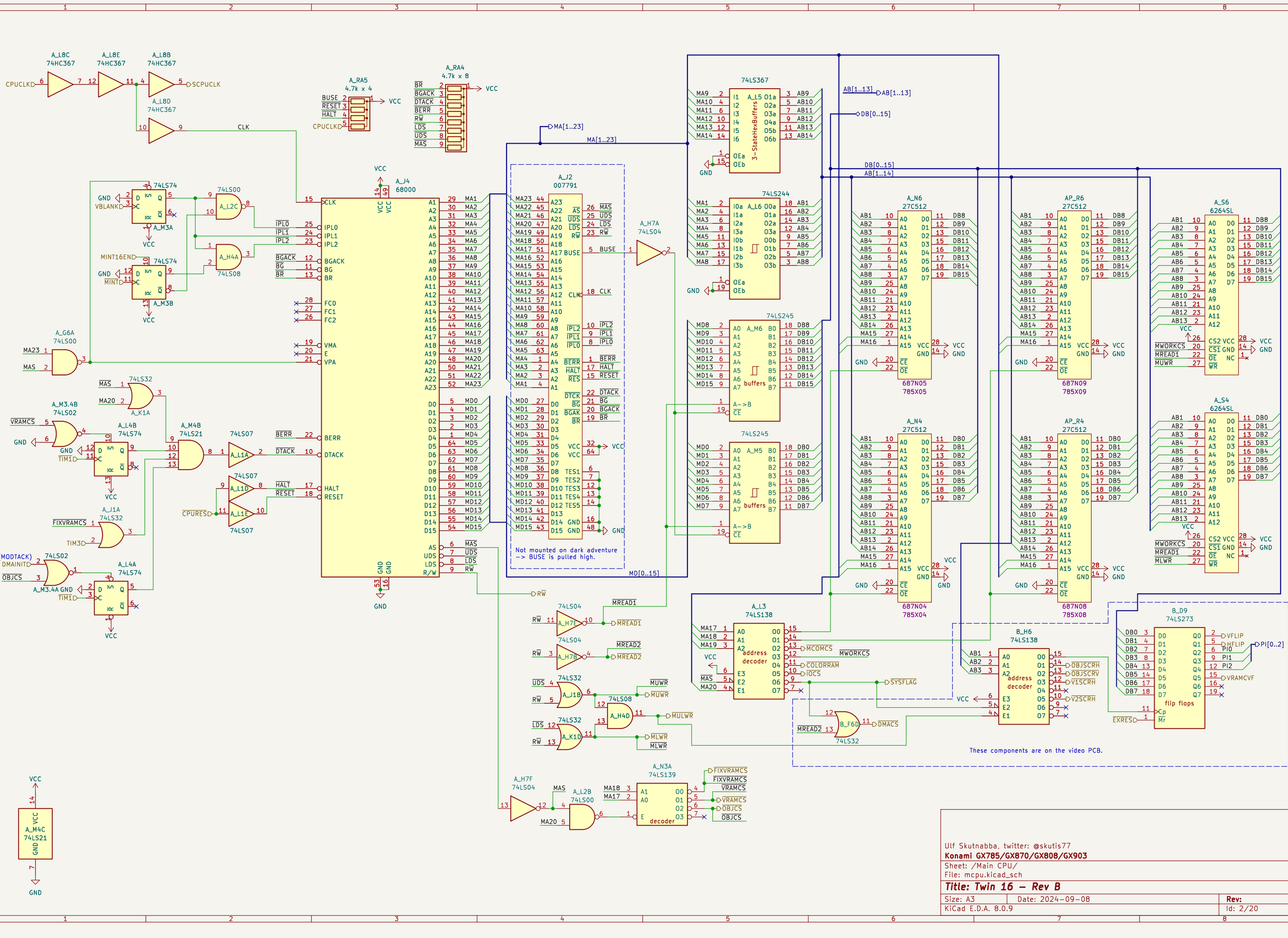
Ulf Skutnabba, twitter: @skutis77

Konami GX785/GX870/GX808/GX903

Sheet: /
File: twin16b.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	Rev:
KiCad E.D.A. 8.0.9		Id: 1/20



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Konami GX785/GX870/GX808/GX903

Sheet: /Main CPU/

File: mcpu.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.9		Id: 2/20

Only used on PWB(A)302190B
Goes to CN2:24.
On PWB(A)302190A CN2:24 is connected to GND.

In the original schematics
DMAINT, MODTACK and DMAINIT are the same

8R and 10R are not mounted when
the GX687 board is used for gradius 2
with GX785 roms.

Ulf Skutnabba, twitter: @skutis77

Konami GX785/GX870/GX808/GX903

Sheet: /Slave CPU/
File: scpu.kicad_sch

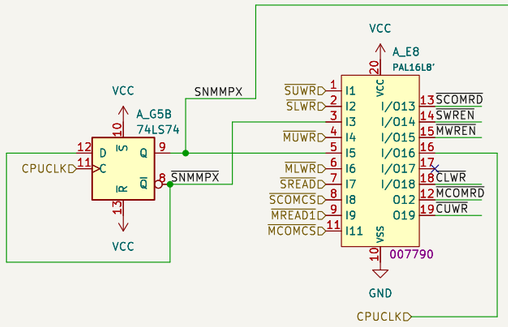
Title: Twin 16 – Rev B

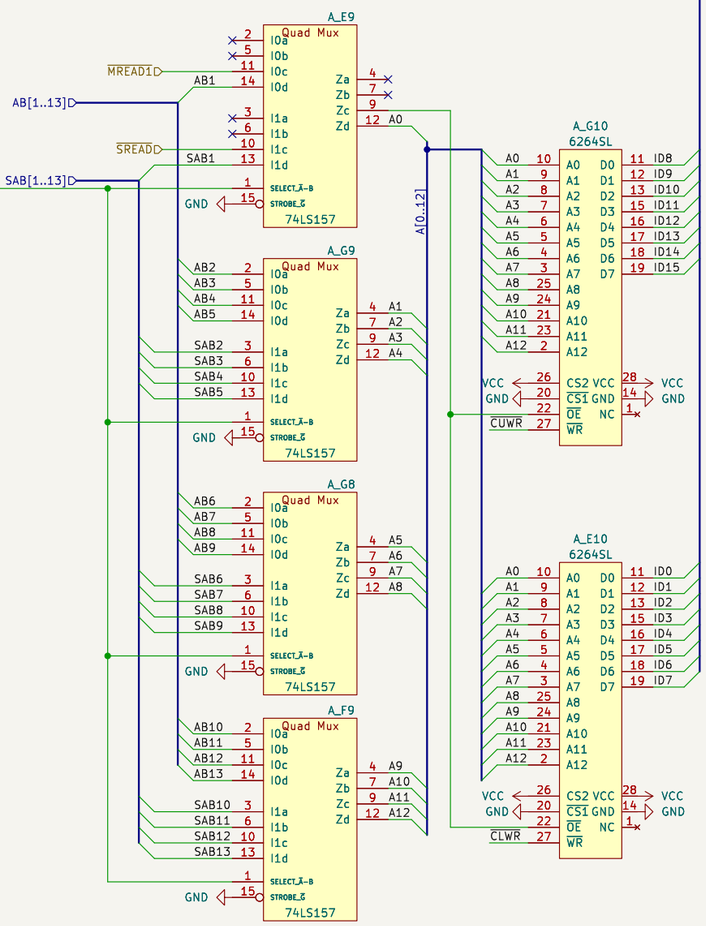
Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.9		Id: 3/20

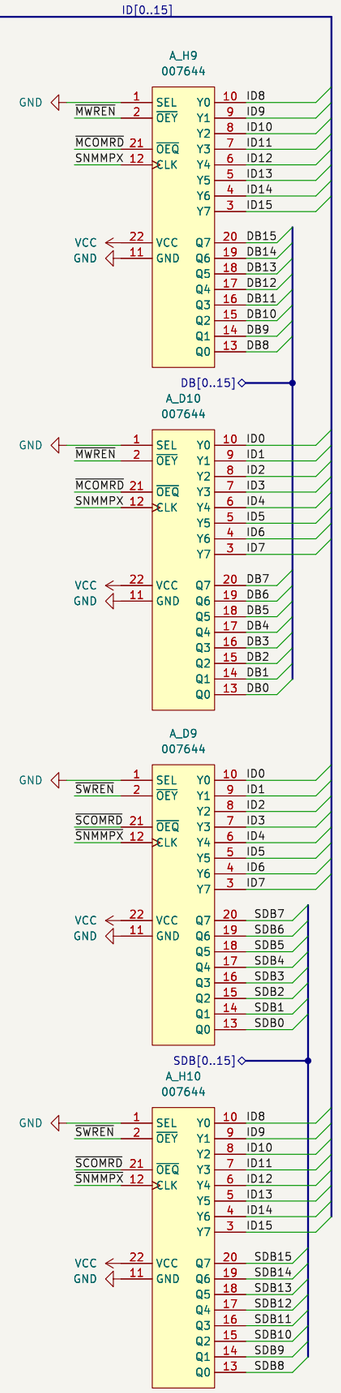
Konami 007790

$\overline{MCOMRD}$ = ~(~$\overline{MREAD1}$ & ~$\overline{MCOMCS}$) = $\overline{MREAD1}$ | $\overline{MCOMCS}$
$\overline{SCOMRD}$ = ~(~$\overline{SREAD}$ & ~$\overline{SCOMCS}$)
$\overline{SWREN}$ = ~(~SNMMPX & $\overline{SREAD}$ & ~$\overline{SCOMCS}$)
$\overline{MWREN}$ = ~(~SNMMPX & $\overline{MREAD1}$ & ~$\overline{MCOMCS}$)
$\overline{CLWR}$ = ~(~$\overline{SLWR}$ & ~$\overline{SNMMPX}$ & SNMMPX & $\overline{SREAD}$ & ~$\overline{SCOMCS}$ & ~CPUCLK | ~SNMMPX & ~$\overline{MLWR}$ & $\overline{MREAD1}$ & ~$\overline{MCOMCS}$ & ~CPUCLK)
$\overline{CUWR}$ = ~(~$\overline{SUWR}$ & ~$\overline{SNMMPX}$ & SNMMPX & $\overline{SREAD}$ & ~$\overline{SCOMCS}$ & ~CPUCLK | ~$\overline{MUWR}$ & ~SNMMPX & $\overline{MREAD1}$ & ~$\overline{MCOMCS}$ & ~CPUCLK)

All AND logic can be converted to OR form. But since AND logic and inverters are used in the PAL16L8 device it is kept in that form.
The equations follow verilog syntax. The active low signals are not inverted, the bar is there only to show activeness.







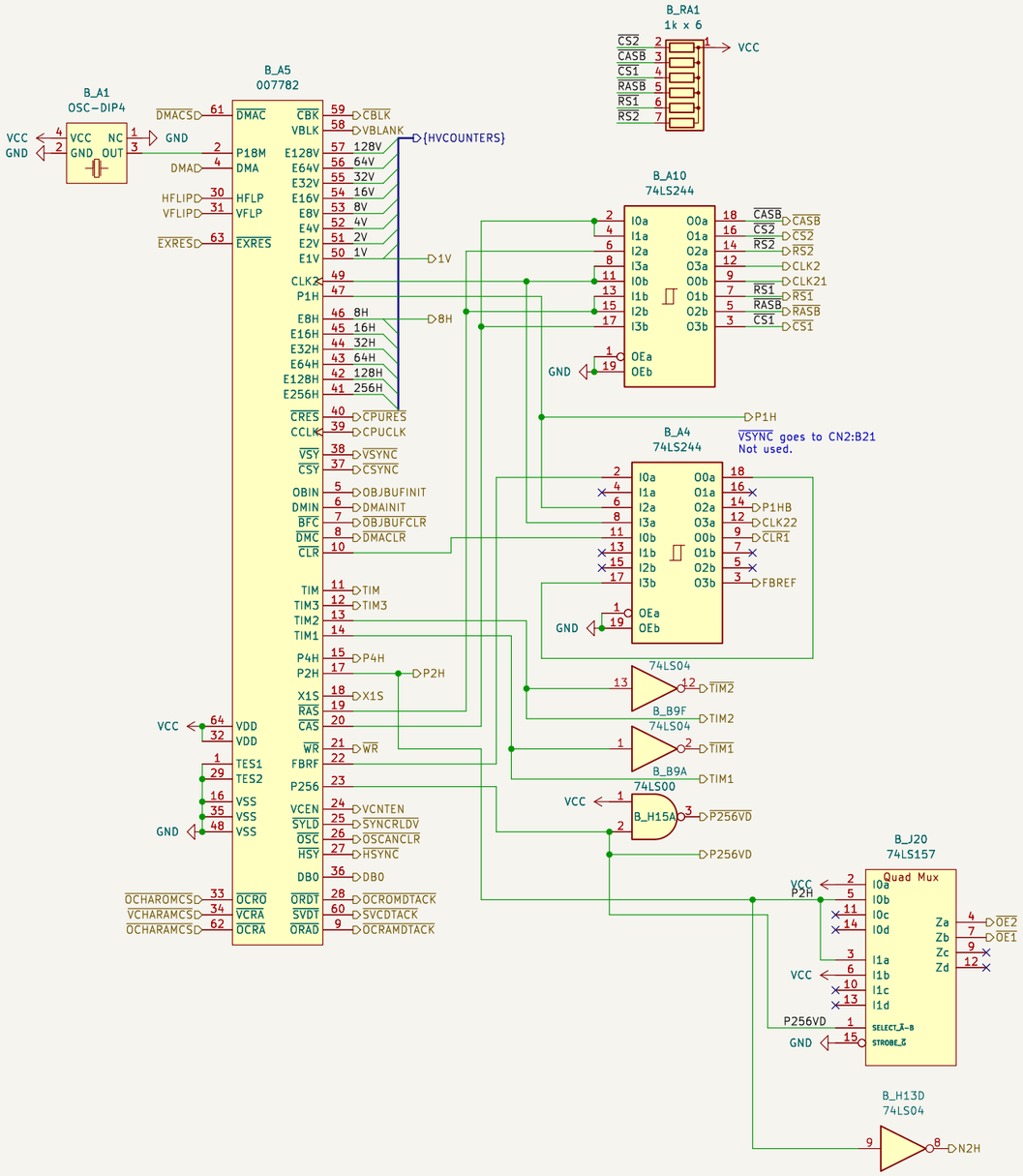
This is from jammarcade.net

Pin 1 = SEL – changes whether the Q outputs require a rising edge clock or not.
Pin 2 = OEY – Output enable for the Y outputs else they are inputs. Active LOW
Pin 21 = OEQ – Output enable for the Q (clocked) outputs else they are inputs. Active LOW
Pin 12 = CLK – Clock input

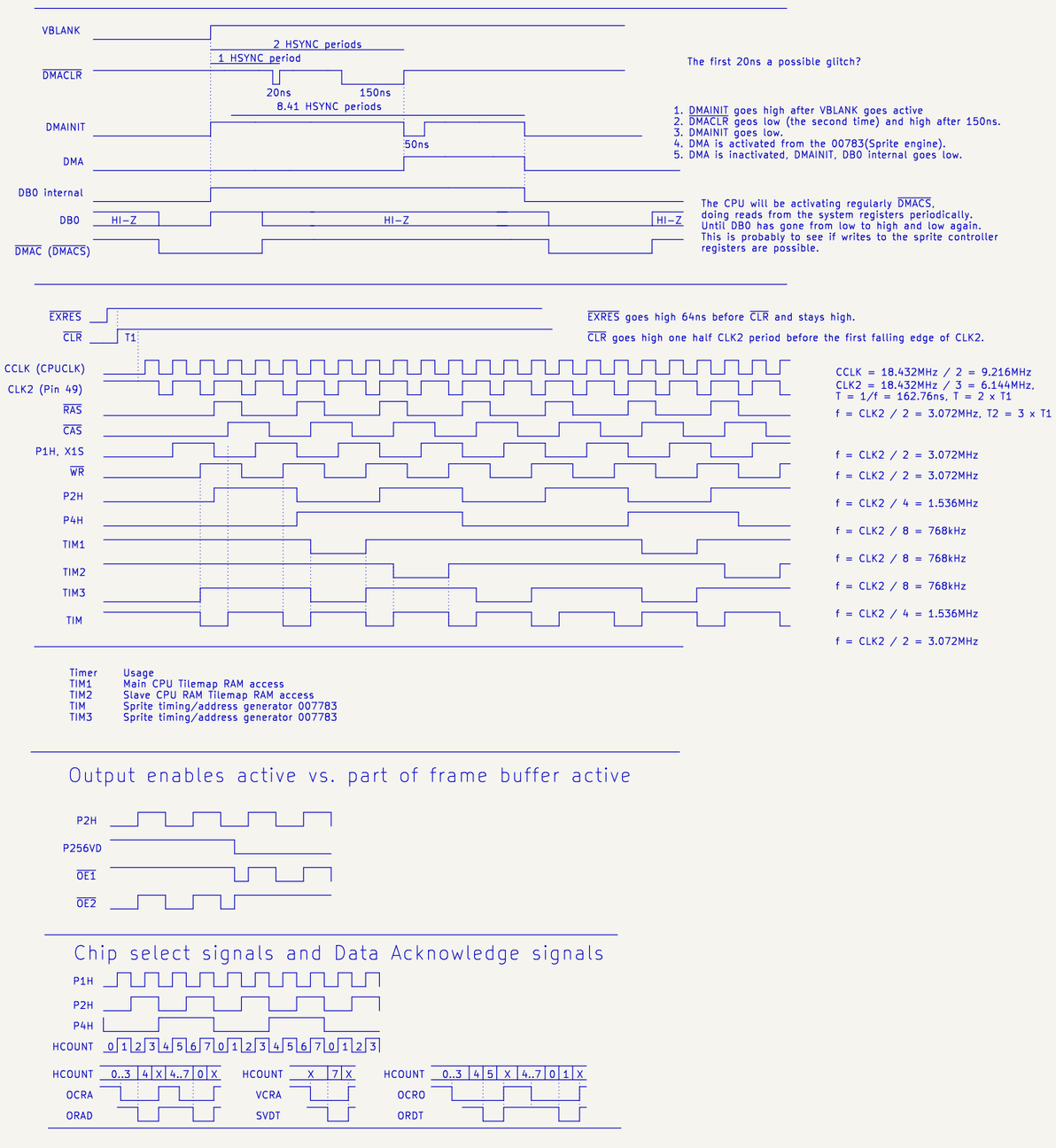
Pins 3–10 = Y outputs – normal outputs. If OEY is LOW these pins become outputs
mirroring the state of the Q input pins. SEL and CLK pins are not used in this mode.

Pins 13–20 = Q outputs – clocked outputs. If OEQ is LOW these pins become outputs.
If SEL is HIGH and CLK is HIGH these outputs will mirror the state of the Y inputs.
If SEL is LOW then the outputs will only change state on a rising edge CLK.

Ulf Skutnabba, twitter: @skutis77
Konami GX785/GX870/GX808/GX903
Sheet: /Shared Resources/
File: shared_resources.kicad_sch
Title: Twin 16 – Rev B
Size: A3 | Date: 2024-09-08 | **Rev:**
KiCad E.D.A. 8.0.9 | Id: 4/20



DMA Synchronization



Timer	Usage
TIM1	Main CPU Tilemap RAM access
TIM2	Slave CPU RAM Tilemap RAM access
TIM	Sprite timing/address generator 007783
TIM3	Sprite timing/address generator 007783

Output enables active vs. part of frame buffer active

Chip select signals and Data Acknowledge signals

Ulf Skutnabba, twitter: @skutis77

Konami GX785/GX870/GX808/GX903

Sheet: /Timing/
File: timing.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.9		Id: 5/20

B_B3 74LS157 Quad Mux
B_B4 74LS157 Quad Mux
B_B5 74LS157 Quad Mux
B_B6 74LS157 Quad Mux
B_B7 74LS273 flip flops
B_B8 74LS157 Quad Mux
B_C3 6264SL
B_C4 6264SL
B_C5 74LS174 hex flip flop
B_C6 74LS174 hex flip flop
B_C9 74LS273 flip flops
B_C10 74LS273 flip flops
B_D5 74LS174 hex flip flop
B_D6 74LS174 hex flip flop
B_D8 27C128 785G14 870F14 808E14 903E14
B_D10 74LS273 flip flops
B_E4A 74LS27
B_E4B 74LS27
B_F6A 74LS32
B_F7 74LS245 buffers
B_F8 74LS245 buffers
B_H13A 74LS04
B_H13F 74LS04
B_A11A 74LS86
B_A11B 74LS86
B_A11C 74LS86
B_A11D 74LS86
B_B10A 74LS86
B_B10B 74LS86
B_B10C 74LS86
B_B10D 74LS86
B_B11A 74LS86
B_B11C 74LS86
B_B11D 74LS86

7ns delay
20ns delay
7ns delay
10ns delay

Palette, HFLIP and VFLIP is valid for eight pixels at a time and becomes valid on the rising edge of P4H

The total delay for a pixel is seven pixel clocks from when P2H and OE goes high and low respectively.

Two pixels color data is output on each rising edge of P1HB. The current pixel data is multiplexed with P1HB (and HFLIP).

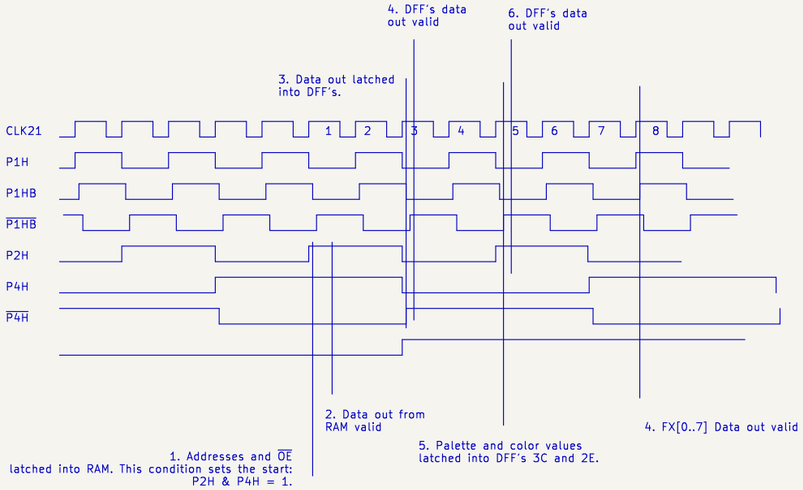
50ns delay from OE active to output data valid

FX[0..3]: Color
FX[4..7]: Palette

RAM
A[0..5] Horizontal position 0..504 in steps of 8
A[6..10] Vertical position 0..248 in steps of 8
A[11.12] Tilemaps? Up to 4.

ROM
A[0..1] Horizontal position 0..6 in steps of 2
A[2..4] Vertical position 0..7
A[5..13] Tile number

D[0..8]: Tile number
D[9..12]: Palette
D13: HFLIP
D14: VFLIP



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Konami GX785/GX870/GX808/GX903

Sheet: /Fixed Layer/
File: fixed_layer.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.9		Id: 6/20

This is from jammarcade.net

Konami 007644

Pin 1 = SEL – changes whether the Q outputs require a rising edge clock or not.
Pin 2 = OEY – Output enable for the Y outputs else they are inputs. Active LOW
Pin 21 = OEQ – Output enable for the Q (clocked) outputs else they are inputs. Active LOW
Pin 12 = CLK – Clock input

Pins 3–10 = Y outputs – normal outputs. If OEY is LOW these pins become outputs mirroring the state of the Q input pins. SEL and CLK pins are not used in this mode.

Pins 13–20 = Q outputs – clocked outputs. If OEQ is LOW these pins become outputs. If SEL is HIGH and CLK is HIGH these outputs will mirror the state of the Y inputs.
If SEL is LOW then the outputs will only change state on a rising edge CLK.

The 007781 is a shift register. All counters related to all timing signals are inside the chip. It is synchronized on the rising edge of CLR and CLK2 after that. It reads 32 bits / 4 bits / pixel = 8 pixels at a time.

1C[0:3]: Layer 1 Pixel color code
1C[4:6]: Layer 1 Palette
2C[0:3]: Layer 2 Pixel color code
2C[4:6]: Layer 2 Palette

VD[0..12]: Tile number
VD[13..15]: Palette Color

Data latched from the RAM:
Edge Tilemap
T1–T2 1
T2–T3 2

It is output 2 pixel clocks later to the ROMs.

When VA is switched to output the bus is updated for each P1H cycle.

The vertical counters are not the same value when they are offset with the scroll registers.

Tilemap 1 and 2
VA[5..0]: Horizontal position: 0..63 -> 0..63 x 8 = 0..504
VA[11..6]: Vertical position: 0..63 -> 0..63 x 8 = 0..504
VA12 = 0: Tilemap 1, VA12 = 1: Tilemap 2

Outputs of RAM active when P4H is low, i.e during T1 and T2.

SYNCRLDV: Usually 60.604Hz
T is ~150ns.

VCNTEN: 8.0000kHz

HSYNC: 16.0000kHz, T2: 5.2175us

Ulf Skutnabba, twitter: @skutis77

Konami GX785/GX870/GX808/GX903

Sheet: /Tile Layers/
File: tile_layers.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.9		Id: 7/20

B_B12
007783

B_C13
6264SL

B_C14
6264SL

B_C12
007784

B_F12
007785

74LS245

74LS157
Quad Mux
B_J13

74LS157
Quad Mux
B_J14

74LS157
Quad Mux
B_J18

74LS157
Quad Mux
B_J16

74LS157
Quad Mux
B_J17

B_F6B
74LS32

B_K12A
74LS32

B_H12C
74LS32

B_H12A
74LS32

B_P11A
74LS74

B_A16
PAL16L8'

B_RA3
2.2k x 8

B_RA2
2.2k x 8

AB[1..13]

SAB[1..14]

OA[0..12]

CRW[0..3]

DMA

{RW1}

OD[0..15]

DB[0..15]

SDB[0..15]

{RW2}

SA[15..20]

O[0..19]

VD[7..0]

HD[8..2]

HJ[0..1]

CX[0..3]

C[0..3]

SPA[0..19]

Each unparsed sprite (object) occupies 40 words in object RAM. Only the first 8 words of each object data is used. The memory range for unparsed data is up 17FFh. That leaves room for 153 (153 x 40 −1 = 17E7h) objects per frame.

On PWB(A) 302190A CHAPAGE (CN2:A23) is connected to GND

007784 Sprite parser

− subtracts the scroll value for the sprites from current coordinates for the sprites.
− When DMA goes high. The 007784 will read 8 words, if bit OD15 of the first word is high then BEFLAG goes high and stays high until the sprite RAM has processed the first four words and possibly updated the four following words at the high end of the RAM. It will compress the eigth words into four words and also subtract the x− and y−scroll values from the x and y coordinates. It will write the four words at the high end of the sprite RAM.
− Addresses at the low end goes from 0x0000..0x0007 and the following cycle at 0x0028..0x002F..

Scroll Register
OBJSCRV Sprite horizontal scroll
OBJSCRH Sprite vertical scroll

Data is latched in the scroll registers when the signals OBJSCRV and OBJSCRH go low.

007793

CLK21′ = CLK21

LRW = -(DMAONS & OBJBUFINIT & CLK21′ |
BEFLAG & DMA & P1H & DMAONS & CLK21′ & (~8H & HFLIP | 8H & ~HFLIP)
~LWR & ~DMAINIT)

URW = -(DMAONS & OBJBUFINIT & CLK21′ |
BEFLAG & DMA & P1H & DMAONS & CLK21′ & (8H & ~HFLIP | ~8H & HFLIP) |
~UWR & ~DMAINIT)

The sprite memory is controlled in three separate cases:
1. OBJBUFINIT
2. BEFLAG & DMA
3. ~DMAINIT

Ulf Skutnabba, twitter: @skutis77

Konami GX785/GX870/GX808/GX903

Sheet: /Sprite Table/
File: sprite_table.kicad_sch

Title: Twin 16 − Rev B

Size: A3	Date: 2024−09−08	**Rev:**
KiCad E.D.A. 8.0.9		Id: 8/20

Graphics ROM and RAM

ROM naming

	ID	P13	P15	P16	P18
cuebrickj	GX903	NP	NP	NP	NP
miaj	GX808	808D15	NP	808D17	NP
hpuncher	GX870	870C15	870C16	870C17	870C18
gradius2 / vulcan	GX785	785F15	785F16	785F17	785F18

NP = Not Populated
On some boards P13,P15,P16 and P18 are NP and replaced with PWB(C) 402145A. See separate schematics.

This is from jammarcade.net

Pin 1 = SEL – changes whether the Q outputs require a rising edge clock or not.
Pin 2 = OEY – Output enable for the Y outputs else they are inputs. Active LOW
Pin 21 = OEQ – Output enable for the Q (clocked) outputs else they are inputs. Active LOW
Pin 12 = CLK – Clock input

Pins 3–10 = Y outputs – normal outputs. If OEY is LOW these pins become outputs mirroring the state of the Q input pins. SEL and CLK pins are not used in this mode.

Pins 13–20 = Q outputs – clocked outputs. If OEQ is LOW these pins become outputs. If SEL is HIGH and CLK is HIGH these outputs will mirror the state of the Y inputs. If SEL is LOW then the outputs will only change state on a rising edge CLK.

Sprite Character Banks

Page	SPA19	SPA18	SPA17
1	X	L	L
2	X	L	H
3	L	H	L
4	H	H	H
5 RAM	X	H	H

Pixel color data output rate is:
16-bits / 2 pixels = 8 bits / pixel.

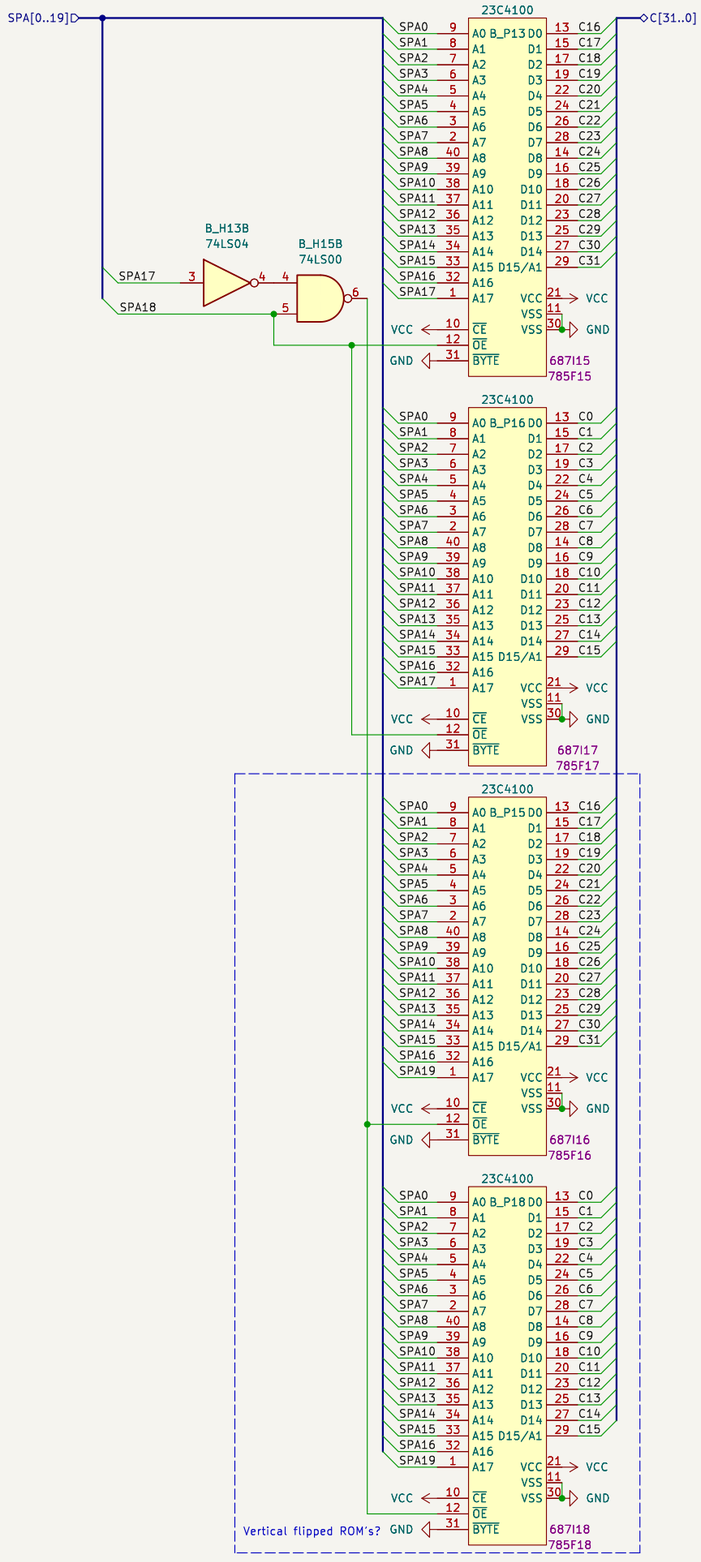
W[15]: Vertical flip

Konami 007788

The inverted ~P2H is written as N2H.

$\overline{OCRDL}$ = ~(~$\overline{SREAD}$ & SAB1 & ~$\overline{OCHARAMCS}$ & $\overline{OCHAROMCS}$ | ~$\overline{SREAD}$ & SAB1 & ~$\overline{OCHAROMCS}$)
$\overline{OCWL}$ = ~(~P2H & $\overline{SREAD}$ & SAB1 & ~$\overline{OCHARAMCS}$)
$\overline{OCRDH}$ = ~(~$\overline{SREAD}$ & ~SAB1 & ~$\overline{OCHARAMCS}$ & $\overline{OCHAROMCS}$ | ~$\overline{SREAD}$ & ~SAB1 & ~$\overline{OCHAROMCS}$)
$\overline{OCWH}$ = ~(~P2H & $\overline{SREAD}$ & ~SAB1 & ~$\overline{OCHARAMCS}$)
$\overline{VCWENL}$ = ~(P2H & P4H & ~$\overline{VCHARAMCS}$ & $\overline{SREAD}$ & SAB1)
$\overline{VCRDL}$ = ~(~$\overline{VCHARAMCS}$ & ~$\overline{SREAD}$ & SAB1)
$\overline{VCRDH}$ = ~(~$\overline{VCHARAMCS}$ & ~$\overline{SREAD}$ & ~SAB1)
$\overline{VCWENH}$ = ~(P2H & P4H & ~$\overline{VCHARAMCS}$ & $\overline{SREAD}$ & ~SAB1)
$\overline{P2HP4H}$ = ~(P2H & P4H)
$\overline{N2HP4H}$ = ~(~P2H & P4H)

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Konami GX785/GX870/GX808/GX903
Sheet: /Sprite RAM/
File: sprite_ram.kicad_sch
Title: Twin 16 – Rev B
Size: A3 | Date: 2024-09-08 | **Rev:**
KiCad E.D.A. 8.0.9 | Id: 9/20



Sheet: /Sprite ROM 1/
File: sprite_rom_1.kicad_sch

Title: Twin 16 – Rev B

Size: A3 | Date: 2024-09-08 | **Rev:**
KiCad E.D.A. 8.0.9 | Id: 10/20

C_A14
4.7k x 8
C'0 C'1 C'2 C'3 C'4 C'5 C'6 C'7
VCC

C'[31..0]

C_A15
C'0 C'1 C'2 C'3 C'4 C'5 C'6 C'7
I0a I1a I2a I3a I0b I1b I2b I3b
O0a O1a O2a O3a O0b O1b O2b O3b
C0 C1 C2 C3 C4 C5 C6 C7
OEa OEb
74LS244
VCC GND

C[31..0]

SPA[19..0]

C_C12A
74LS08
SPA17 SPA18

CR_D8
4.7k x 8
SPA7 SPA6 SPA5 SPA4 SPA3 SPA2 SPA1 SPA0
VCC

C_D9
SPA0 SPA1 SPA2 SPA3 SPA4 SPA5 SPA6 SPA7
I0a I1a I2a I3a I0b I1b I2b I3b
O0a O1a O2a O3a O0b O1b O2b O3b
SPA'0 SPA'1 SPA'2 SPA'3 SPA'4 SPA'5 SPA'6 SPA'7
GND OEa OEb
74LS244

C_D10
4.7k x 8
C'8 C'9 C'10 C'11 C'12 C'13 C'14 C'15
VCC

C_D11
C'8 C'9 C'10 C'11 C'12 C'13 C'14 C'15
I0a I1a I2a I3a I0b I1b I2b I3b
O0a O1a O2a O3a O0b O1b O2b O3b
C8 C9 C10 C11 C12 C13 C14 C15
OEa OEb
74LS244
VCC GND

C_C8
4.7k x 8
SPA9 SPA10 SPA8 SPA11 SPA12 SPA15 SPA13 SPA14
VCC

C_D9
SPA8 SPA9 SPA10 SPA11 SPA12 SPA13 SPA14 SPA15
I0a I1a I2a I3a I0b I1b I2b I3b
O0a O1a O2a O3a O0b O1b O2b O3b
SPA'8 SPA'9 SPA'10 SPA'11 SPA'12 SPA'13 SPA'14 SPA'15
GND OEa OEb
74LS244

SPA'[15..0]

C_C10
4.7k x 8
C'16 C'17 C'18 C'19 C'20 C'21 C'22 C'23
VCC

C_C11
C'19 C'17 C'18 C'16 C'20 C'21 C'22 C'23
I0a I1a I2a I3a I0b I1b I2b I3b
O0a O1a O2a O3a O0b O1b O2b O3b
C19 C17 C18 C16 C20 C21 C22 C23
OEa OEb
74LS244
VCC GND

Mirrored when connected to PWB(B)

C_CN3
SPA9 a1 b1 SPA10
SPA8 a2 b2 SPA11
SPA7 a3 b3 SPA12
SPA6 a4 b4 SPA13
SPA5 a5 b5 SPA14
SPA4 a6 b6 SPA15
SPA3 a7 b7 SPA16
SPA2 a8 b8 SPA17
SPA1 a9 b9 SPA18
SPA0 a10 b10 SPA19
GND a11 b11 VCC
a12 b12
a13 b13
a14 b14
a15 b15
a16 b16
C15 a17 b17 C31
C14 a18 b18 C30
C13 a19 b19 C29
C12 a20 b20 C28
C11 a21 b21 C27
C10 a22 b22 C26
C9 a23 b23 C25
C8 a24 b24 C24
C7 a25 b25 C23
C6 a26 b26 C22
C5 a27 b27 C21
C4 a28 b28 C20
C3 a29 b29 C19
C2 a30 b30 C18
C1 a31 b31 C17
C0 a32 b32 C16
DIN64PS

C_B14
4.7k x 8
C'24 C'25 C'26 C'27 C'28 C'29 C'30 C'31
VCC

C_B15
C'24 C'25 C'26 C'27 C'28 C'29 C'30 C'31
I0a I1a I2a I3a I0b I1b I2b I3b
O0a O1a O2a O3a O0b O1b O2b O3b
C31 C30 C29 C28 C27 C26 C25 C24
OEa OEb
74LS244
VCC GND

$\overline{SPOE}$[7..0]

C_D12
4.7k x 4
SPA16 SPA17 SPA18
VCC

C_D13
74LS138
SPA16 SPA17 SPA18
A0 A1 A2
address decoder
E3 E2 E1
VCC GND
O0 O1 O2 O3 O4 O5 O6 O7
$\overline{SPOE0}$ $\overline{SPOE1}$ $\overline{SPOE2}$ $\overline{SPOE3}$

SPA19

C_C15B
74LS32
$\overline{SPOE4}$

C_C13A
74LS04

C_C15A
74LS32
$\overline{SPOE5}$

C_C15C
74LS32
$\overline{SPOE7}$

C_C15D
74LS32
$\overline{SPOE6}$

C_C13B
74LS04

C_C13C
74LS04

C_C13D
74LS04

C_C13E
74LS04

C_C13F
74LS04

C_C12B
74LS08

C_C12C
74LS08

C_C12D
74LS08

PWB(C) 402145A
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Sheet: /Sprite ROM logic/
File: sprite_rom_logic.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.9		Id: 11/20

C_B12 C_B11 C_B10 C_B9 C_A12 C_A11 C_A10 C_A9
C_B8 C_B7 C_B6 C_B5 C_B4 C_B3 C_B2 C_B1
C_D8 C_D7 C_D6 C_D5 C_D4 C_D3 C_D2 C_D1
C_A8 C_A7 C_A6 C_A5 C_A4 C_A3 C_A2 C_A1
27C512
F17B F17D F17F F17H F18B F18D F18F F18H
F17A F17C F17E F17G F18A F18C F18E F18G
F15B F15D F15F F15H F16B F16D F16F F16H
F15A F15C F15E F15G F16A F16C F16E F16G
SPA'0 SPA'1 SPA'2 SPA'3 SPA'4 SPA'5 SPA'6 SPA'7 SPA'8 SPA'9 SPA'10 SPA'11 SPA'12 SPA'13 SPA'14 SPA'15
A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12 A13 A14 A15
D0 D1 D2 D3 D4 D5 D6 D7
C'0 C'1 C'2 C'3 C'4 C'5 C'6 C'7 C'8 C'9 C'10 C'11 C'12 C'13 C'14 C'15 C'16 C'17 C'18 C'19 C'20 C'21 C'22 C'23 C'24 C'25 C'26 C'27 C'28 C'29 C'30 C'31
VCC GND CE OE
SPOE0 SPOE1 SPOE2 SPOE3 SPOE4 SPOE5 SPOE6 SPOE7
On the PCB the position is D9
SPOE[7..0]
SPA'[15..0]
C'[31..0]

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Sheet: /Sprite ROM 2/
File: sprite_rom_2.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.9		Id: 11/20

Frame buffer 1

Frame buffer 2

Not certain which of the parts 6KB,6UB and 6UD are used where. Some errors in the original schematics.

Framebuffer 1 is outputting pixel data when P256VD is low and writes to the FB is possible when P256VD is high. The conditions are reveresed for FB2.

Writes to FB are possible when the "pixel part" is sleected from the mux.
If this expression is true
(R[0:3] != 4'h0) && (FBWM1 | FBWM2 | -WR | -VPD8)
data is written to DRAM.

Data is written when:
– WR is active.
– Not a transparent pixel (4'h0).
– FBWM1 and FBWM2 are not active: No writes to frame buffer going on.
– VPD8 (vertical counter highest bit, VCNT > 256) is not active. Outside visible area.

Frame buffer 1 Palette color

Frame buffer 2 Palette color

OCD[0:3]: Pixel color
OCD[4:7]: Palette

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Sheet: /Sprite Frame Buffers/
File: sprite_frame_buffers.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	Rev:
KiCad E.D.A. 8.0.9		Id: 13/20

Components are on Video PCB

Components are on Main PCB

the four bits of each layer has the following meaning:
0000: Transparent
1111: Shadow
Else: Opaque

Konami 007789

i11 (1C6) and i5 (PI2) is not used

~o15 = OCD0 & OCD1 & OCD2 & OCD3 | ~OCD0 & ~OCD1 & ~OCD2 & ~OCD3

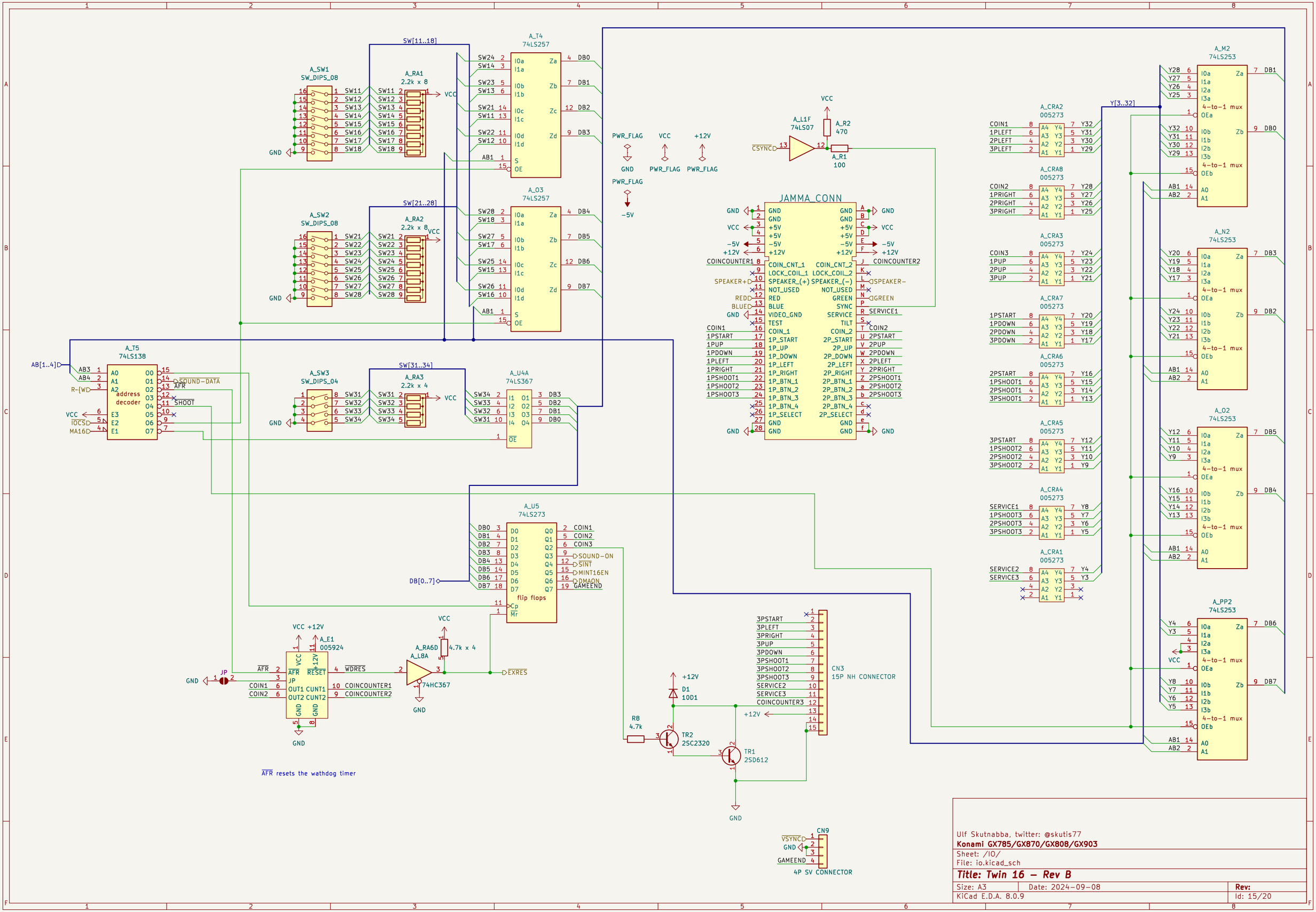
o16 = OCD0 & OCD1 & OCD2 & OCD3

SELA = 1C |
PI0 & ~PI1 & FX |
~PI1 & OCD0 & OCD1 & OCD2 & OCD3 & FX |
~PI1 & ~OCD0 & ~OCD1 & ~OCD2 & ~OCD3 & FX |
PI1 & ~2C0 & OCD0 & OCD1 & OCD2 & OCD3 & ~2C1 & ~2C2 & ~2C3 |
PI1 & ~2C0 & ~OCD0 & ~OCD1 & ~OCD2 & ~OCD3 & ~2C1 & ~2C2 & ~2C3

~SELB = ~1C & OCD0 & OCD1 & OCD2 & OCD3 |
~1C & ~OCD0 & ~OCD1 & ~OCD2 & ~OCD3 |
~1C & PI0 & ~PI1 & FX |
~1C & PI0 & PI1 & 2C0 & BUS |
~1C & PI0 & PI1 & BUS & 2C1 |
~1C & PI0 & PI1 & BUS & 2C2 |
~1C & PI0 & PI1 & BUS & 2C3

~SHADOW = ~1C & ~PI1 & OCD0 & OCD1 & OCD2 & OCD3 |
~1C & ~PI0 & OCD0 & OCD1 & OCD2 & OCD3 & ~FX |
~1C & PI0 & OCD0 & OCD1 & OCD2 & OCD3 & ~BUS |
~1C & PI0 & ~2C0 & OCD0 & OCD1 & OCD2 & OCD3 & ~2C1 & ~2C2 & ~2C3

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Sheet: /Color mixer/
File: color_mixer.kicad_sch
Title: Twin 16 – Rev B
Size: A3 | Date: 2024-09-08 | **Rev:**
KiCad E.D.A. 8.0.9 | Id: 14/20



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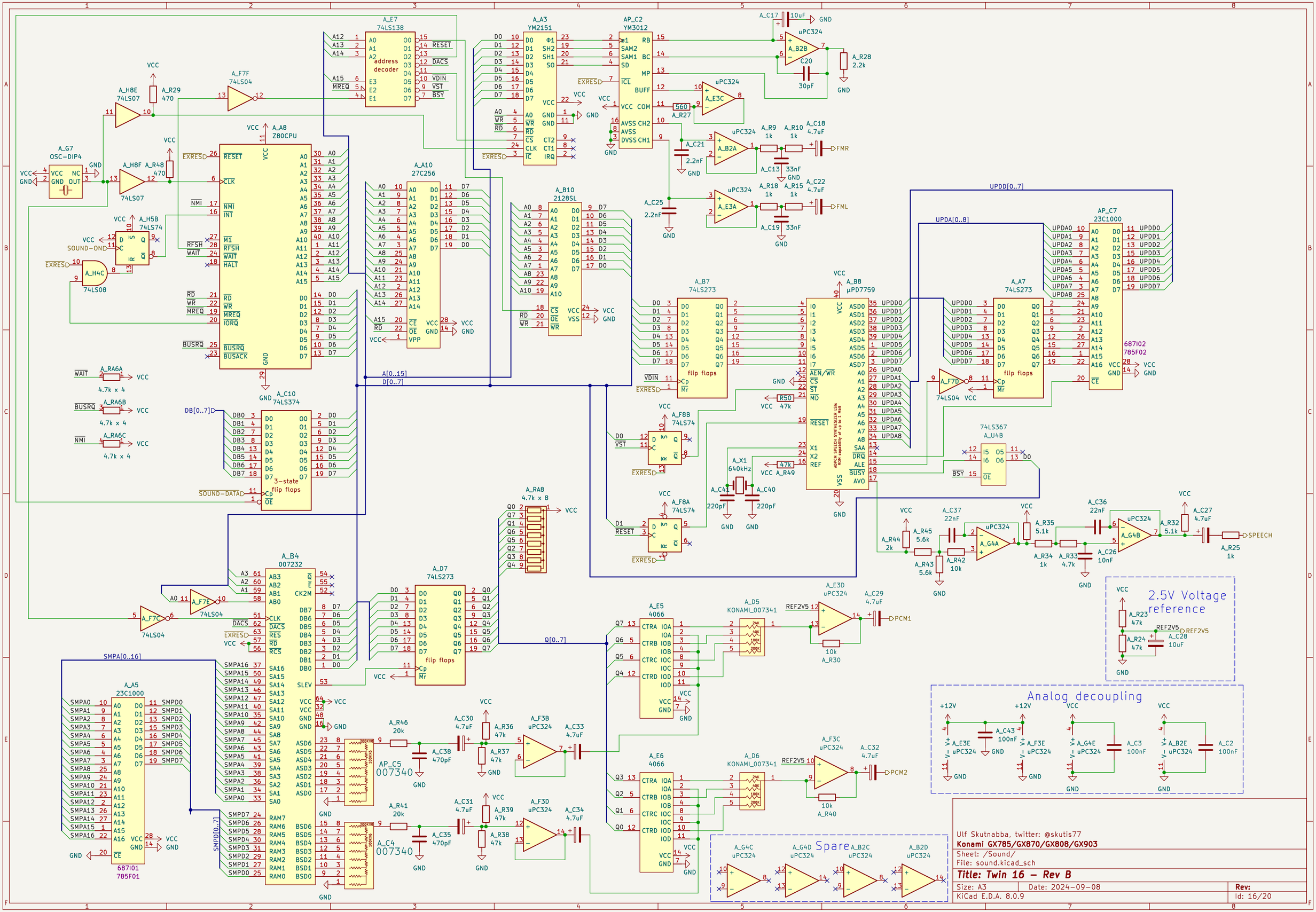
Konami GX785/GX870/GX808/GX903

Sheet: /IO/

File: io.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.9		Id: 15/20


A_E7
74LS138
address decoder
A_A3
YM2151
AP_C2
YM3012
A_A8
Z80CPU
A_A10
27C256
A_B10
2128SL
A_B7
74LS273
A_B8
µPD7759
ADPCM SPEECH SYNTHESIZER LSI
ROM capability of up to 1 Mbit
A_A7
74LS273
AP_C7
23C1000
687I02
785F02
A_G7
OSC-DIP4
A_X1
640kHz
A_H8E
74LS07
A_F7F
74LS04
A_H8F
74LS07
A_F7E
74LS04
A_F7C
74LS04
A_F7D
74LS04
A_H5B
74LS74
A_H4C
74LS08
A_F8B
74LS74
A_F8A
74LS74
A_C10
74LS374
3-state flip flops
A_B4
007232
A_D7
74LS273
A_A5
23C1000
687I01
785F01
AP_C5
007340
A_C4
007340
A_RA8
4.7k x 8
A_RA6A
4.7k x 4
A_RA6B
4.7k x 4
A_RA6C
4.7k x 4
A_E3C
uPC324
A_B2A
uPC324
A_E3A
uPC324
A_B2B
uPC324
A_E3D
uPC324
A_F3C
uPC324
A_F3B
uPC324
A_G4A
uPC324
A_G4B
uPC324
A_E5
4066
A_E6
4066
A_D5
KONAMI_007341
A_D6
KONAMI_007341
74LS367
A_U4B
A_G4C
uPC324
A_G4D
uPC324
A_B2C
uPC324
A_B2D
uPC324
Spare
2.5V Voltage reference
Analog decoupling
A[0..15]
D[0..7]
SMPA[0..16]
SMPD[0..7]
DB[0..7]
Q[0..7]
UPDA[0..8]
UPDD[0..7]
VCC
GND
A_R29 470
A_R48 470
A_R27 560
A_R28 2.2k
A_R50 47k
A_R49 47k
A_R46 20k
A_R41 20k
A_R36 47k
A_R37 47k
A_R39 47k
A_R38 47k
A_R30 10k
A_R40 10k
A_R9 1k
A_R10 1k
A_R18 1k
A_R15 1k
A_R45 5.6k
A_R43 5.6k
A_R44 2k
A_R42 10k
A_R35 5.1k
A_R34 1k
A_R33 4.7k
A_R32 5.1k
A_R25 1k
A_R23 47k
A_R24 47k
A_C17 10uF
A_C20 30pF
A_C21 2.2nF
A_C25 2.2nF
A_C13 33nF
A_C19 33nF
A_C18 4.7uF
A_C22 4.7uF
A_C27 4.7uF
A_C29 4.7uF
A_C30 4.7uF
A_C31 4.7uF
A_C32 4.7uF
A_C33 4.7uF
A_C34 4.7uF
A_C38 470pF
A_C35 470pF
A_C41 220pF
A_C40 220pF
A_C36 22nF
A_C37 22nF
A_C26 10nF
A_C43 100nF
A_C3 100nF
A_C2 100nF
A_C28 10uF
FMR
FML
PCM1
PCM2
SPEECH
REF2V5
SOUND-ON
SOUND-DATA
EXRES
WAIT
BUSRQ
NMI
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Sheet: /Sound/
File: sound.kicad_sch
Title: Twin 16 – Rev B
Size: A3
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KiCad E.D.A. 8.0.9
Rev:
Id: 16/20

To enable stereo sound output on connector CN8:
Use the connector which shortens Pin 1 -> Pin 2 and
Pin 3 -> Pin 4 on Connector CN7.

FMRD
A_R17 1k
SPEECHD
A_R12 7.5k
A_R11 5.1k
PCM1D
A_R19 5.1k
PCM2D
REF2V5D
A_E3B uPC324
A_R13 12k
+12V
A_R52 3.3k
A_C15 4.7uF
A_R14 10k
CN6 4P NH CONN
GND
CN7 4P NH CONN
APOT_R53 1k
APOT_R52 1k
A_C11 100nF
A_C12 100nF
FMLD
A_R16 1k
A_R22 7.5k
A_R31 5.1k
A_R20 5.1k
A_F3A uPC324
A_R21 12k
A_R53 3.3k
A_C16 4.7uF
A_R26 10k
A_C9 100uF
ACAP_C10 100uF
A_D1 MB3722
IN-A
IN+A
IN+A
IN-B
VCC
GNDIN
GNDOUT
BYPASS
BSA
OUTA
BSB
OUTB
ACAP_C3 1000uF
A_C8 100uF
A_C7 100uF
A_C14 10uF
ACAP_C2 1000uF
A_C6 100nF
A_R6 1
A_C5 100nF
A_R7 1
A_C1 1000uF
SPEAKER-
SPEAKER+
CN8 4P NH CONN

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Konami GX785/GX870/GX808/GX903

Sheet: /Sound mixer/
File: sound_mixer.kicad_sch

Title: Twin 16 - Rev B

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.9		Id: 17/20

D_F1A 74LS04 · D_G1B 74LS32 · D_F1B 74LS04 · D_G1A 74LS32 · D_E1 74LS174 · D_G1C 74LS32 · D_F1C 74LS04 · D_F2D 74LS32 · D_E2A 74LS32

MA16' · IOCS' · RAM · AB'10 · CPURES' · RWEN · DB''12 · HRAMEN · LRAMEN

DB''[12..8]

D_G2 6264SL · D_I2 6264SL · D_H2 6264SL · D_J2 6264SL

AB'1 AB'2 AB'3 AB'4 AB'5 AB'6 AB'7 AB'8 AB'9 DB''8 DB''9 DB''10 DB''11 VCC

DB'0 DB'1 DB'2 DB'3 DB'4 DB'5 DB'6 DB'7 · DB'8 DB'9 DB'10 DB'11 DB'12 DB'13 DB'14 DB'15

CS2 VCC CS1 GND OE NC WR

RAMEN · MLWR' · MUWR' · HRAMEN · MREAD2' · LRAMEN

D_F2C · D_F2B · D_F2A · D_E2D · D_E2C · D_E2B · 74LS32

DB'[15..0] · AB'[10..1]

74LS245 buffers A->B CE

D_C1: DB0 DB1 DB2 DB3 DB4 DB5 DB6 DB7 → DB'0 DB'1 DB'2 DB'3 DB'4 DB'5 DB'6 DB'7

D_D1: DB8 DB9 DB10 DB11 DB12 DB13 DB14 DB15 → DB'8 DB'9 DB'10 DB'11 DB'12 DB'13 DB'14 DB'15

D_C2: AB1 AB2 AB3 AB4 AB5 AB6 AB7 AB8 → AB'1 AB'2 AB'3 AB'4 AB'5 AB'6 AB'7 AB'8

D_D2: AB9 AB10 MREAD2 MLWR MUWR MA16 IOCS CPURES → AB'9 AB'10 MREAD2' MLWR' MUWR' MA16' IOCS' CPURES'

AB[10..1] · DB[15..0]

CN1 Conn_02x20_Odd_Even

GND · DB0 DB1 DB2 DB3 DB4 DB5 DB6 DB7 DB8 DB9 DB10 DB11 DB12 DB13 DB14 DB15 · AB1 AB2 AB3 AB4 AB5 AB6 AB7 AB8 AB9 AB10 · MREAD2 MWR IOCS CPUCLKD VCC · MLWR MA16 CPURES VBLANK VCC · (Not used) (Not used)

CN5 is on main board.
CN1 is on sub board PWB451803 and mates to CN5.

D_F1D 74LS04 · D_G1D 74LS32 · D_F1E 74LS04 · D_F1F 74LS04

VMEM · PWR_FLAG

VCC VMEM · D_U1 M51944BSL · D_R1 10k · RAMEN

If VCC > 4.25V OUT is activated

D_D11 1S1588 · D_D12 1S1588 · D_BT1 3V

VMEM is connected to VCC on the RAM IC's.

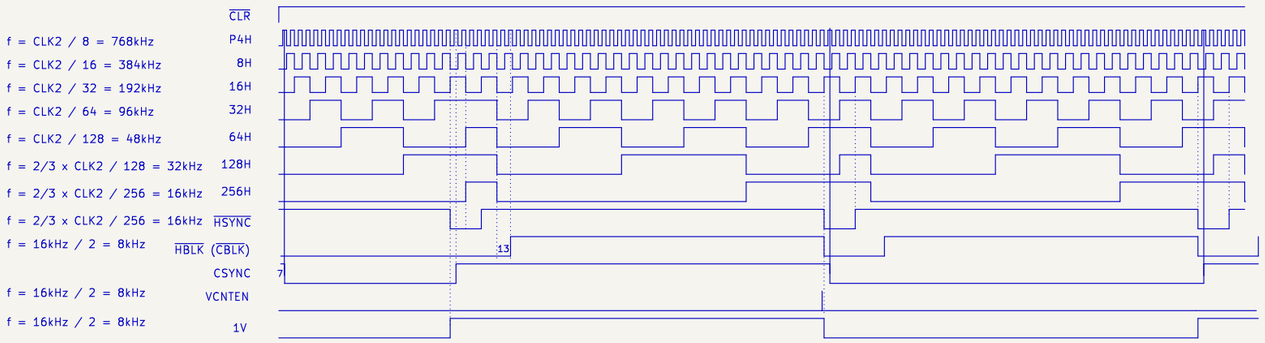
GC903
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Konami GX785/GX870/GX808/GX903

Sheet: /Battery Powered RAM/
File: bpram.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.9		Id: 18/20

Horizontal signals

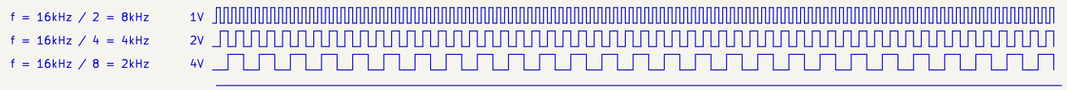


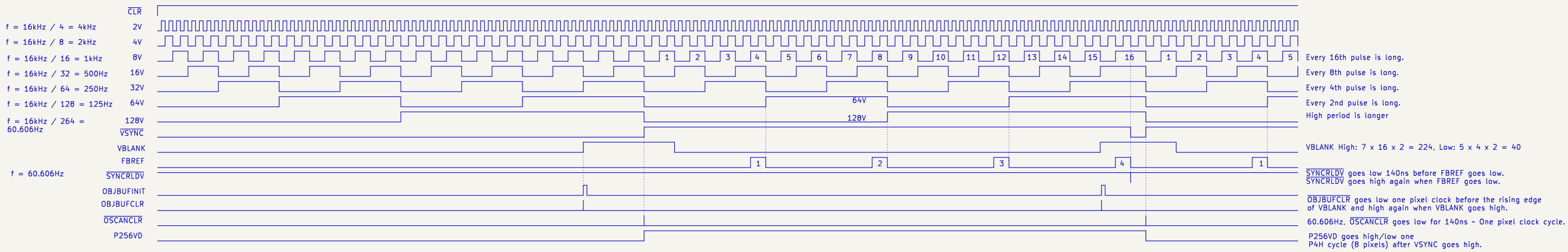
– The first VCNTEN is skipped after reset.
It goes low 140ns before HSYNC goes low,
and high again when HSYNC goes low.
VCNTEN is active right before every second falling edge of HSYNC.

– CPURES goes high, and stays high, on the seventh falling edge of HSYNC.

The first HSYNC is at HCOUNT 175 after reset.

Vertical signals





FBREF is measured at pin 22. At U186:3 FBREF is delayed by 22ns going high and 12ns going low.

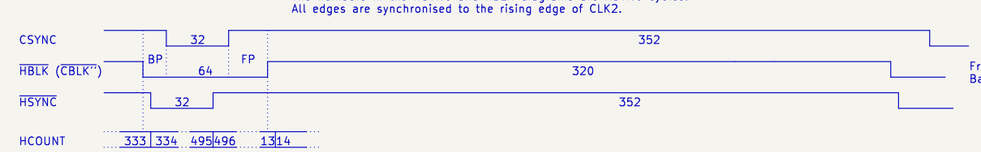
Horizontal and vertical synch timing diagrams

Pixel clock CLK2 = 18.432MHz / 3 = 6.144MHz

Horizontal lines
HSYNC freq = 6.144MHz / 384 = 16kHz
HSYNC period = 1 / 16kHz = 62.5us
Pixels / line: 384
Active pixels: 320
Blank pixels: 64

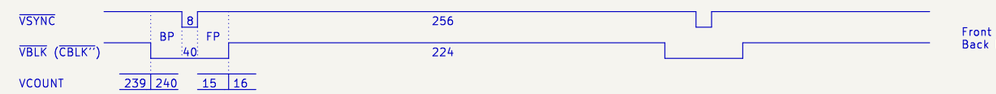
Vertical lines
VSYNC freq = 16kHz / 264 = 60.606060Hz
VSYNC period = 1 / f = 264 / 16kHz = 16.5ms
scanlines / frame: 264
Active lines: 224
Blank lines: 40

The numbers in the HSYNC and HBLK diagrams are HSYNC cycles.
All edges are synchronised to the rising edge of CLK2.



Front Porch (FP) 24 px,
Back Porch (BP) 8 px

The numbers in the VSYNC and VBLK diagrams are HSYNC cycles.
All edges are synchronised to the falling edge of HSYNC.



Front Porch (FP) and
Back Porch (BP) 16 HSYNC cycles.

HCOUNT is bits [256H, 128H, 64H, 32H, 16H, 8H, P4H, P2H, P1H]
VCOUNT is bits [128V, 64V, 32V, 16V, 8V, 4V, 2V, 1V]

CBLK'' is at the output of color mixer 007593.
CBLK' is shifted inside the 007593 one pixel clock.

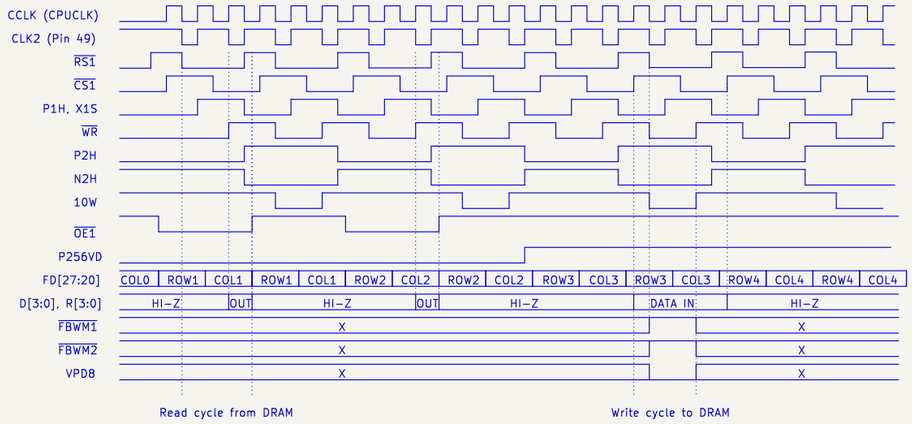
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Sheet: /Timing diagrams/
File: timing_diagrams.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.9		Id: 19/20



The same row and column addresses are repeated twice.
One for read cycle and the other for write cycles.

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Konami GX785/GX870/GX808/GX903

Sheet: /Sprite timing diagrams/
File: sprite_timing_diagrams.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.9		Id: 20/20